CS 4375 HW3 — Programming Part: Results

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Dataset: UCI Predict Students' Dropout and Academic Success (4,424 rows, 36 features, 3 classes). Split 70/15/15 stratified (3096/664/664), seed 42, 100 epochs max, batch size 64, early stopping (patience 15). Accuracy and macro-F1 are in %. "Best ep." is the epoch with the lowest validation loss.

All 24 configurations (training and validation only)

#	Activation	Hidden	Optimizer	LR	Best ep.	Train acc	Val acc	Val macro-F1
1	sigmoid	64	sgd	0.1	52	79.07	77.11	70.13
2	sigmoid	64	sgd	0.01	100	77.07	76.05	66.65
3	sigmoid	64	adam	0.01	5	80.43	76.81	69.01
4	sigmoid	64	adam	0.001	24	80.43	76.20	68.13
5	sigmoid	64-32	sgd	0.1	91	80.17	77.11	70.99
6	sigmoid	64-32	sgd	0.01	100	74.35	73.19	52.85
7	sigmoid	64-32	adam	0.01	6	80.98	77.11	70.45
8	sigmoid	64-32	adam	0.001	27	81.04	76.66	69.66
9	tanh	64	sgd	0.1	14	79.88	76.36	68.50
10	tanh	64	sgd	0.01	97	79.43	77.41	69.95
11	tanh	64	adam	0.01	2	78.23	76.51	68.26
12	tanh	64	adam	0.001	10	80.56	75.75	68.46
13	tanh	64-32	sgd	0.1	6	78.42	76.36	68.57
14	tanh	64-32	sgd	0.01	47	79.49	76.66	69.38
15	tanh	64-32	adam	0.01	4	82.75	77.11	69.52
16	tanh	64-32	adam	0.001	6	80.68	76.51	69.38
17	relu	64	sgd	0.1	14	82.24	76.66	69.94
18	relu	64	sgd	0.01	97	80.81	77.56	70.57
19	relu	64	adam	0.01	1	79.20	74.70	65.10
20	relu	64	adam	0.001	10	83.04	76.20	69.47
21	relu	64-32	sgd	0.1	6	80.56	76.36	68.51
22	relu	64-32	sgd	0.01	57	81.56	76.51	68.79
23	relu	64-32	adam	0.01	2	81.49	75.45	66.85
24	relu	64-32	adam	0.001	6	82.95	76.20	68.47

Best configuration per activation (chosen by validation macro-F1)

Activation	Hidden	Optimizer	LR	Val acc	Val macro-F1	Test acc	Test macro-F1
sigmoid	64-32	sgd	0.1	77.11	70.99	79.67	73.09
tanh	64	sgd	0.01	77.41	69.95	79.07	71.46
relu	64	sgd	0.01	77.56	70.57	78.61	70.43
Logistic regression baseline (same split)						78.16	71.35

Learning curves (best configuration of each activation)

